

Hadi Zanganeh Kia

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Professional Summary

Postdoctoral researcher in Atmospheric Sciences with expertise in atmospheric predictability, numerical weather prediction, and quantitative analysis using large-scale gridded datasets. Experienced in Python-based scientific computing, machine learning for weather/climate forecasting, and high-performance computing on supercomputers. Strong background in moisture dynamics, precipitation recycling, and developing novel atmospheric research methodologies using multi-model frameworks.

Education

2022–2025: PhD, Atmospheric Sciences, University of Houston, GPA: 3.95

2016–2019: MSc, Mechanical Engineering (Energy Conversion), University of Tehran, GPA: 3.77

2011–2015: BSc, Mechanical Engineering, K.N. Toosi University of Technology, GPA: 3.24

Dissertation

Title: Computational Fluid Dynamics and Machine Learning Applications in Atmospheric Science: From Microscale Particle Transport to Mesoscale Cloud Forecasting

Advisor: Dr. Yunsoo Choi (ychoi6@uh.edu)

Description: Developed high-resolution CFD simulations and deep learning models for cloud cover prediction using multi-satellite observations, gridded climate reanalysis data (ERA5), and physics-informed neural networks.

Professional Experience

2025–Present: Postdoctoral Research Fellow, University of Houston

Developing advanced numerical methods for atmospheric moisture tracking and precipitation recycling dynamics. Conducting large-scale climate simulations on NCAR Derecho supercomputer using WRF and iCESM frameworks. Processing and analyzing multi-terabyte ERA5/CFSR climate datasets to investigate sub-seasonal precipitation predictability and regional water cycle dynamics.

2025: Teaching Assistant, University of Houston

Instructed students on HPC systems (PBS, SLURM), Linux environments, and scientific Python for large-scale data processing and visualization.

2022–2025: Research Assistant, University of Houston

Conducted atmospheric modeling and developed comprehensive Python tools for quantitative analysis of climate data. Contributed to peer-reviewed publications and contributed to research proposals.

2018–2021: MATLAB Tutor, Iran University of Science and Technology

Taught MATLAB programming, Simulink, and scientific data visualization.

2017–2018: Research Assistant, VFERI

Analyzed metro system development and transportation dynamics.

Research Projects

- Atmospheric moisture tracking using Two-Layer Dynamic Recycling Model (2L-DRM) and WRF-WVT framework for sub-seasonal precipitation predictability

- Numerical weather prediction: Multi-year WRF simulations (1979–2014) for Western US hydro-climate analysis
- Cloud cover forecasting using Graph Attention Networks with CERES satellite data and gridded reanalysis inputs
- Physics-informed deep learning for atmospheric predictability and cloud parametrization
- Large-scale climate data processing and analysis on NCAR Derecho supercomputer (multi-terabyte datasets)
- LES of atmospheric convection and boundary layer dynamics using OpenFOAM
- Tropical cyclone moisture analysis and atmospheric river transport (WRF-WVT)
- Renewable energy: Wind turbine CFD and aerodynamic optimization in Ansys Fluent
- CFD of rotor dynamics and industrial fluid mechanics applications

Publications

1. Hadi Zanganeh Kia, Yunsoo Choi, Rashik Islam, Reetam Dimri, and Shihab Ahmad Shahriar. (2025). Short-Term Forecasting of Total Cloud Cover over the United States and the Texas Gulf Coast with Graph-Based Deep Learning. *Earth Systems and Environment*, 1-22.
2. Momeni, M., Kashfi Yeganeh, A., Zanganeh Kia, H., Ghahremanloo, M., et al. (2025). Using multi-satellite observations to constrain ammonia emissions and unlock their potential over open water. *Scientific Reports* 15(1), 26784.
3. Shahriar, Shihab Ahmad, Choi, Yunsoo, Islam, Rashik, Zanganeh Kia, Hadi, Shams, Seyedeh Reyhaneh, and Salman, Ahmed Khan. (2025). Forecasting PM2.5 and Assessing Health Impacts in Texas Using Advanced Deep Learning Models. *Earth Systems and Environment*, 1-22.
4. Hadi Zanganeh Kia, Fan Yang, Marat Khairoutdinov, Raymond A. Shaw, Aaron Wang, and Yunsoo Choi. (2025). Exploring the impact of surface topography on Rayleigh-Bénard dry convection in the Pi cloud chamber using OpenFOAM: In cylindrical and rectangular geometries. *Atmospheric Research*, 108144.
5. Aaron Wang, Silvio Schmalfuß, Kamal Kant Chandrakar, Hadi Zanganeh Kia, Fan Yang, Mikhail Ovchinnikov, Raymond A. Shaw, and Yunsoo Choi. (2025). An intercomparison of wall fluxes in a turbulent thermal convection chamber: Direct numerical simulations and wall-modeled large-eddy simulations enhanced by machine learning. *Physics of Fluids* 37(4).
6. Hadi Zanganeh Kia. (2025). Computational Fluid Dynamics and Machine Learning Applications in Atmospheric Science: From Microscale Particle Transport to Mesoscale Cloud Forecasting. Doctoral Dissertation, University of Houston.
7. Hadi Zanganeh Kia, Yunsoo Choi, Delaney Nelson, Jincheol Park, and Arman Pouyaei. (2023). Investigating the Impact of Microclimate on Airborne Particle Dispersion, Including Sneeze Pathogens, in a Poorly Ventilated Urban Outdoor Environment. AGU23.
8. Hadi Zanganeh Kia, Yunsoo Choi, Delaney Nelson, Jincheol Park, and Arman Pouyaei. (2023). Large eddy simulation of sneeze plumes and particles in a poorly ventilated outdoor air condition: A case study of the University of Houston main campus. *Science of The Total Environment* 891, 164694.
9. Ranjbar, MH, Zanganeh Kia, H., Nasrazadani, SA, Gharali, K., Nathwani, J. (2020). Experimental and numerical investigations of actuator disks for wind turbines. *Energy Science & Engineering* 8(7), 2371-2386.

10. Ranjbar, M., Nasrazadani, S., Zanganeh Kia, H., Gharali, K. (2019). Reaching the Betz limit experimentally and numerically. *Energy Equipment and Systems* 7(3), 271-278.

Technical Skills

- **Python:** NumPy, Pandas, Xarray, Dask (large-scale data processing)
- **ML/DL:** PyTorch, TensorFlow, scikit-learn, Graph Neural Networks
- **NWP/Climate:** WRF, iCESM, CESM
- **Data:** ERA5, gridded reanalysis, satellite (CERES), observational data
- **HPC:** NCAR Derecho, PBS/SLURM, parallel computing, MPI
- **CFD:** OpenFOAM, Ansys Fluent, mesh generation
- **CAD/Meshing:** SolidWorks, ICEM, Ansys Meshing
- **Tools:** LaTeX, Git, MATLAB, Bash, CDO/NCO

Languages

Persian: Native

English: Fluent

French: Beginner

Awards & Distinctions

- Best First-Year PhD Student Award – UH Earth & Atmospheric Sciences
- Graduate Research Grant – UH Earth & Atmospheric Sciences
- Ranked 320th / 120,000 in National University Entrance Exam (Iran)
- Full Scholarship – K.N. Toosi University of Technology
- Ranked 1st – Computer Science Olympiad

References

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- **Dr. Raymond A. Shaw**

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